Close CRM Pipeline Build Guide

STEP-BY-STEP IMPLEMENTATION MANUAL

AWS Region: us-east-2
Architecture: Medallion

PHASE 1: INGESTION & STORAGE SETUP

STEP 1 Create S3 Buckets for Data Layers

Create two S3 buckets in AWS to hold raw and transformed data:

- **Bronze Bucket:** `insightflow-bronze-bucket-sd` (stores raw webhook JSON files).
- **Gold Bucket:** `insightflow-gold-bucket-sd` (stores processed, business-ready leads).

STEP 2 Set Up SQS Delay Queue

Create an SQS Standard Queue to hold incoming lead notifications for 10 minutes so Close CRM can finish populating lead owner details:

- **Name:** `crm-lead-delay-queue`
- **Delivery Delay:** Set to `600 seconds` (10 minutes).

STEP 3 Deploy Ingestion Lambda & API Gateway

Build an endpoint to receive live webhooks from Close CRM and store them in S3:

- Create an Ingestion Lambda function that writes incoming webhook JSON into the Bronze S3 bucket and sends a message to the SQS queue.
- Expose the Lambda via **API Gateway** with a `/leads` POST endpoint.
- Copy your live Invoke URL (e.g., `https://mbdf9v87n7.execute-api.us-east-2.amazonaws.com/leads`).

STEP 4 Register Webhook in Close CRM

Connect Close CRM to your AWS endpoint using the Close API:

- Send a POST request to `https://api.close.com/api/v1/webhook/subscriber/` using your Close API Key.
- Set `url` to your API Gateway `/leads` endpoint and subscribe to the `lead.created` event.
- Save the returned `signature_key` to verify webhook HMAC signatures.

PHASE 2: ENRICHMENT WORKER & TESTING

STEP 5 Deploy Enrichment Lambda Worker

Create a second Lambda function that triggers when messages exit the 10-minute SQS queue:

- Set SQS `crm-lead-delay-queue` as the event trigger for this Lambda.
- Configure an Environment Variable: `GOLD_BUCKET_NAME` = `insightflow-gold-bucket-sd`.
- Write code to read the raw Bronze JSON, parse nested fields under `event.data`, and save the enriched record into the Gold bucket.

STEP 6 Test End-to-End Pipeline

Trigger a live event and verify data movement through all layers:

- Create a new test lead (e.g., "Premier Motors") in the Close CRM UI.
- Verify the raw JSON appears immediately in the Bronze S3 bucket.
- Check SQS to confirm 1 message is in **Delayed** status holding for 10 minutes.

STEP 7 Verify Gold Tier Output

After the 10-minute timer expires, confirm final execution:

- Check CloudWatch logs for `crm-lead-enrichment-worker` to ensure successful execution.
- Open `insightflow-gold-bucket-sd` and verify that `enriched_[lead_id].json` contains the extracted lead name and status.